1. Array & String Patterns

- **Two Pointers** (opposite ends / same direction) — sorted arrays, palindromes, pair sums
- **Sliding Window** (fixed & variable size) — subarrays, substrings, max/min window problems
- **Prefix Sum / Difference Array** — range sum queries, subarray sum equals K
- **Kadane's Algorithm** — max subarray sum
- **Fast & Slow Pointers (Floyd's Cycle Detection)** — cycle detection in arrays/linked lists
- **Merge Intervals** — overlapping intervals, scheduling
- **Cyclic Sort** — problems with numbers in range [1, n]
- **In-place reversal** (arrays/linked lists)

2. Searching & Sorting Patterns

- **Modified Binary Search** — rotated arrays, search in 2D matrix, peak element
- **Binary Search on Answer** — "minimize the maximum" / "find smallest k such that..." problems
- **Quickselect** — Kth largest/smallest element
- **Merge Sort / Quick Sort variants** — inversion count, custom sort logic

3. Linked List Patterns

- **Reversal of Linked List** (iterative/recursive, in groups of k)
- **Fast & Slow Pointers** — middle node, cycle detection, palindrome check
- **Dummy Node technique**

4. Stack & Queue Patterns

- **Monotonic Stack** — next greater/smaller element, histogram problems
- **Monotonic Deque** — sliding window maximum
- **Stack-based simulation** — valid parentheses, expression evaluation

5. Heap / Priority Queue Patterns

- **Top K Elements**
- **Two Heaps** (min-heap + max-heap) — median of data stream
- **K-way Merge** — merge k sorted lists/arrays

6. Tree Patterns

- **DFS (preorder/inorder/postorder)** — path sum, diameter, LCA
- **BFS / Level Order Traversal**

- **Tree Construction** (from traversals)
- **Binary Search Tree operations**
- **Trie (Prefix Tree)** — word search, autocomplete

7. Graph Patterns

- **BFS / DFS traversal**
- **Topological Sort** (Kahn's / DFS-based) — course schedule, dependency resolution
- **Union-Find (Disjoint Set Union)** — connected components, cycle detection
- **Dijkstra's Algorithm** — shortest path (weighted, non-negative)
- **Bellman-Ford** — shortest path with negative weights
- **Floyd-Warshall** — all-pairs shortest path
- **Minimum Spanning Tree** (Kruskal's / Prim's)
- **Bipartite Graph check**

8. Backtracking Patterns

- **Subsets / Combinations / Permutations**
- **N-Queens style constraint satisfaction**
- **Word Search / Path finding in grid**

9. Dynamic Programming Patterns

- **0/1 Knapsack**
- **Unbounded Knapsack**
- **Longest Common Subsequence (LCS) family**
- **Longest Increasing Subsequence (LIS)**
- **Matrix Chain Multiplication (interval DP)**
- **DP on Grids** (unique paths, min path sum)
- **DP on Trees**
- **Bitmask DP**
- **State machine DP** (buy/sell stock problems)
- **Digit DP**

10. Greedy Patterns

- **Interval scheduling / activity selection**
- **Jump Game type problems**
- **Huffman encoding style problems**

11. Bit Manipulation Patterns

- **XOR tricks** — single number problems

- **Bit masking for subsets**

12. Miscellaneous but Common

- **Sliding Window + HashMap combo** — anagrams, substring problems
- **Matrix traversal patterns** (spiral, rotate, DFS/BFS on grid)
- **Design problems** (LRU Cache, etc. — combines hashmap + linked list)